

2.02 SAMPLE VS CENSUS

To collect information, we usually survey a representative group. This process is called taking a sample.

To collect information about a whole population, *all* people or items must be surveyed. This is called taking a census. People who are members of small groups can be missed in a sample. We always use a census when we want to make sure that the views of small groups are included.

The Australian Bureau of Statistics conducts a national census every five years. The census occurs in each year ending in a 1 or a 6.

Sample	Census
• Surveys a selected group of people or items	• Surveys all people or items in the population
• Gives approximate information about the population	• Gives exact information about the population
• Simple and inexpensive	• Complex and expensive
• Can be done quickly	• Takes a lot of time to collect and process the information

2.03 TYPES OF SAMPLES

If we decide to use a sample for a survey or questionnaire, there are four types of samples we can use.

Random sample

In a random sample, every member of the population has an equal chance of being included. For example, when a computer selects a customer at random to be surveyed about a mobile phone company, every customer of that company has an equal chance of being selected.

Stratified sample

In a stratified sample, different categories in a population are represented according to the size of each category and then members of each category are selected randomly.

For example, a school's population may be made up of 72% junior students and 28% senior students. A stratified sample of school students must also be made up of 72% junior students and 28% senior students.

Systematic sample

In a systematic sample, selections are made on a regular basis. For example, testing every 500th battery manufactured to check that the machines producing the batteries are working properly.

Self-selected sample

In a self-selected sample, whoever wishes to participate answers the questions asked. For example, when a television show asks people to ring in and answer a Yes/No question, anyone can participate.